When Agreement Hides Error: Shared Measurement Influence in MRI Slice Interpolation

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Abstract

Prediction disagreement is widely used as reliability evidence, yet it can be small when estimates share error. We study this failure in deterministic MRI inter-slice interpolation, where the physical measurements supporting one missing target can be controlled. Target-preserving disjoint contexts expose more paired-error energy than clean shared controls. A preregistered fastMRI $\times 4$ falsification extends this geometry from HAT/SwinIR to I3Net, an adapted SAINT-style network, and an internal CycleINR-style coordinate model: observable-error gains are 17.8/17.6/8.0/13.7/22.6 percentage points, and every examination-bootstrap interval excludes zero. Ranking improvement is heterogeneous, including a slightly negative SAINT point estimate and a strong CycleINR effect. An exact analytical ladder rejects anchor count as a sufficient monotone explanation. Local neural finite-response influence remains stable for HAT/SwinIR and SAINT; SAINT shows strong broad association with error alignment ($\rho = .706$), but only a small, uncertain advantage over count and a fixed-count partial $\rho = .063$ [$-.021$,.149]. I3Net and CycleINR fail the frozen amplitude/sign stability gate and are excluded from influence claims. Thus architecture falsification supports a physical-dependence–error-geometry mechanism while falsifying universal influence superiority and ranking benefit. Acquired-anchor cross-validation remains a lightweight consequence rather than a uniquely superior score. Disagreement is informative only to the extent that its generating measurements exert sufficiently independent, stably measurable influence.

1. Introduction

Sparse through-plane MRI is routinely densified by interpolation or learned slice synthesis. A reconstruction can look anatomically plausible while smoothing, displacing, or inventing structure, so a reliability signal is desirable when dense targets are absent. Ensembles, test-time transformations, and cross-context consistency commonly use output variation as such a signal [3, 5, 11]. Their appeal is intuitive: if plausible predictors disagree, at least one deserves scrutiny. Their blind spot is equally fundamental but less often controlled experimentally: predictors can agree because they reuse the same influential information and therefore make aligned errors.

We turn this abstract dependence problem into a physical intervention. For one missing MRI slice, phase-shifted acquisition grids can provide reconstruction contexts that preserve the target while changing which acquired slice anchors support it. This makes it possible to ask not merely whether outputs differ, but whether dependence among their generating measurements determines which target-error components disagreement can observe.

Figure 1 states the proposed chain. For predictions $f_a = y + e_a$ and $f_b = y + e_b$, output separation observes $e_a - e_b$. Shared measurements can induce similar local estimator responses; similar responses can align e_a and e_b; aligned errors cancel in the difference. The elementary identity is not our novelty. The scientific question is what determines the error inner product. Our evidence identifies shared local measurement influence as one determinant in MRI interpolation.

This formulation also explains why a tempting early hypothesis failed. Integer anchor overlap has the right overall trend but is not a monotone dose: one weak shared anchor can matter less than another, and an exact four-anchor ladder reverses at zero-to-one overlap. We therefore replace count by estimator influence. For analytical interpolation this is derived from coefficients. For frozen HAT and SwinIR networks it is approximated by the cosine between output responses to identical finite perturbations of the same physical anchor. The quantity is explicitly local to a target, context, input, and plane.

Our contributions are threefold. First, we establish and decompose *consensus blindness* with same-target, measurement-controlled interventions across three MRI cohorts, then preregister a five-neural-architecture falsification. Second, we show that shared estimator influence refines raw overlap for analytical estimators and stable

HAT/SwinIR/SAINT probes, while retaining the SAINT fixed-count uncertainty, NingXia nonreplication, physical-plane dependence, and I3Net/CycleINR stability failures as boundaries. Third, we show a lightweight acquired-anchor consequence without claiming unique score superiority. The main result is a falsifiable mechanism for when disagreement is trustworthy, not another uncertainty leaderboard.

2. Related Work

MRI reconstruction and uncertainty. Probabilistic, Bayesian, ensemble, and posterior-sampling methods quantify uncertainty in accelerated MRI [3, 10, 12, 14]. Training-free posterior approximation also addresses uncertainty in computational inverse problems [20]. These methods estimate predictive distributions or risk. Hu *et al*. showed that uncertainty variation can miss systematic MRI reconstruction error [8]. We study a narrower complementary question: how dependence between the physical measurements generating two deterministic predictions changes the target error visible in their disagreement.

Diversity, consistency, and dependence. Deep ensembles use predictive diversity for uncertainty [11]; diversity interacts with individual error and generalization rather than replacing them [2, 16]. MRI self-consistency methods such as SPIRiT enforce compatibility with acquired measurements [13]. Our intervention differs from both: we hold the missing target fixed, explicitly manipulate reuse of physical slice measurements, and relate their estimator influence to target-error alignment and observability.

Slice synthesis and acquisition design. SAINT and I3Net learn anisotropic medical slice synthesis [17, 18]; CycleINR uses an implicit coordinate representation with cycle consistency for arbitrary-scale medical volume super-resolution [4]. Single-stack slice-to-volume reconstruction couples interpolation and motion [19]. Leave-one-slice-out interpolation checks predate this work [7]. Shifted multi-slice super-resolution and acquisition-design studies exploit complementary measurements for reconstruction accuracy [6, 15]. We neither introduce a new interpolator nor claim the holdout operation as novel. Our distinction is using same-target physical dependence as a controlled reliability variable and attempting to falsify the mechanism under substantially different interpolation interfaces.

3. From Disagreement to Shared Influence

3.1. Error observability

Let y be the missing target slice and $f_i = y + e_i$ two deterministic reconstructions. Then

$$\|f_a - f_b\|^2 = \|e_a\|^2 + \|e_b\|^2 - 2\langle e_a, e_b\rangle. \tag{1}$$

Writing $\bar{e} = (e_a + e_b)/2$ and $\delta = (e_a - e_b)/2$ gives

$$\tfrac{1}{2}(\|e_a\|^2 + \|e_b\|^2) = \|\bar{e}\|^2 + \|\delta\|^2. \tag{2}$$

We call the normalized terms *shared* and *observable* error fractions. If $e_a = e_b \neq 0$, disagreement is zero despite error; if $e_a = -e_b$, all paired error energy is observable. Equation 1 is a starting identity. The mechanism question is what makes $\langle e_a, e_b\rangle$ large.

The same geometry gives a one-sided statement. By the triangle inequality,

$$\max(\|e_a\|, \|e_b\|) \geq \tfrac{1}{2}\|f_a - f_b\|. \tag{3}$$

Large separation therefore certifies that at least one reconstruction has nonzero error of a corresponding scale. No converse exists: the right side can vanish for arbitrarily large common error. This asymmetry is why we use *observability* rather than calibration terminology. Disagreement may supply evidence of failure, but agreement cannot by itself supply evidence of correctness.

3.2. Analytical and neural influence

For a local linear estimator $f = \sum_j c_j x_j$, embed two contexts in the union of physical anchor positions, with zero coefficients at unused anchors. Overlap count records common nonzero locations but ignores their unequal weights. We use coefficient cosine, signed dot product, and common absolute mass

$$I_{\cap}(a, b) = \sum_j \min(|c_{a,j}|, |c_{b,j}|) \tag{4}$$

as analytical influence summaries.

For a frozen nonlinear interpolator F_c, we perturb the same shared physical anchor j in contexts a and b with an identical seeded field $\epsilon\xi_{js}$. The centered output response is

$$r_{cjs} = F_c(x + \epsilon\xi_{js}) - F_c(x). \tag{5}$$

Neural shared influence is the mean response cosine between r_{ajs} and r_{bjs} over shared anchors and four frozen seeds. The primary perturbation is a centered Rademacher field at 1% of the globally normalized case range. This is a local response approximation, not a global coefficient, attribution theorem, or property of a network independent of target and plane.

4. Experimental Design

4.1. Cohorts and frozen estimators

We use 24 held-out fastMRI brain T2 examinations [9], 115 held-out IXI brain T2 examinations [1], and 22 out-of-fold NingXia knee T2 series from eight patients. NingXia inference resamples patients with all subordinate series retained. Frozen HAT and SwinIR $\times 4$ reconstructions test the original nonlinear influence claim; linear and cubic interpolation

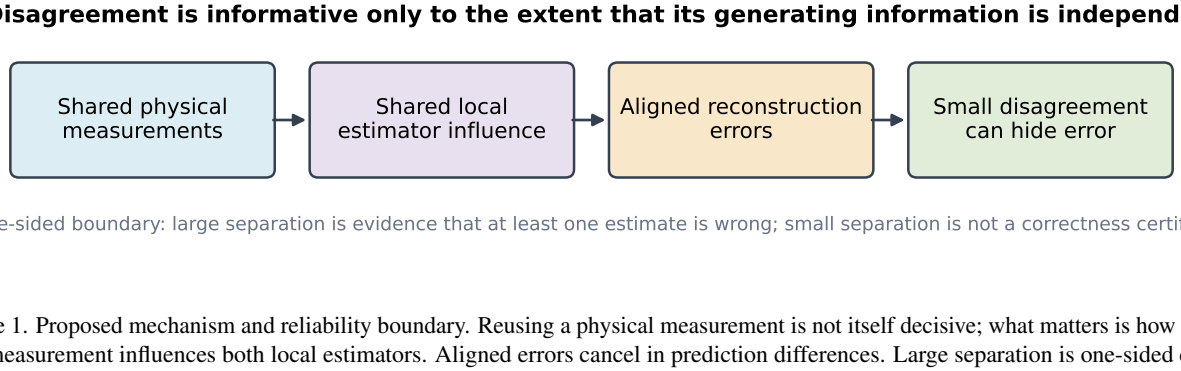


Figure 1. Proposed mechanism and reliability boundary. Reusing a physical measurement is not itself decisive; what matters is how similarly that measurement influences both local estimators. Aligned errors cancel in prediction differences. Large separation is one-sided evidence that at least one estimate is inaccurate, but small separation cannot certify correctness.

provide analytical support. The architecture falsification adds locally trained I3Net, adapted SAINT-style, and internal CycleINR-style checkpoints. They test architecture lineage, not exact official implementations or weights. Their phase-0 wrapper reproduces every stored fastMRI output exactly. fastMRI and IXI also support the frozen retrospective $\times 4 \rightarrow \times 2$ consequence. Dense slices create retrospective anchors and evaluate errors but never enter a reliability score.

4.2. Controlled measurement contexts

The zero-overlap condition reconstructs the same missing plane from phase-0 and phase-2 $\times 4$ lattices. A clean maximal-sharing control applies horizontal flip equivariance to identical anchors while preserving target, anchor identity, z order, distances, count, mask, and preprocessing. The primary endpoint is the Spearman correlation between masked prediction disagreement and paired reconstruction MAE. Intervals use 5,000 examination/patient-cluster bootstrap replicates.

An exact four-anchor ladder fixes target-relative absolute distances $\{1, 2, 3, 4\}$ and compares a reference sign pattern with all valid patterns sharing $k \in \{0, 1, 2, 3, 4\}$ positions. Distance-weighted local linear regression predicts the same target. All valid configurations are averaged at each k; none is selected by outcome. This test asks whether count is a sufficient monotone dose and provides exact coefficients for Eq. 4.

The pair-input neural grid uses a reference context $(-2, +2)$ and eight target-preserving comparators. Target, input count, model, mask, and preprocessing are fixed; span, asymmetry, overlap, and physical identity are recorded. Confirmation includes three physical planes per examination. A cleaner subset keeps four comparators with exactly one shared anchor. For four-anchor I3Net/CycleINR probes, every context remains a regular $\times 4$ grid; a frozen 24-case IXI subset uses one central plane, two overlap-three contexts, and five zero-overlap contexts. Four Rademacher fields at amplitudes $+.005, +.01, +.02, -.01$ define the stability gate ($\rho \geq .90$ for every alternate setting). Models failing any gate remain eligible for the physical-dependence phenomenon but not for influence claims.

4.3. Practical consequence and statistics

Given acquired anchors $x_0, x_4, x_8, \ldots$, Anchor-CV withholds each interior x_{4j}, predicts it linearly from $x_{4(j-1)}$ and $x_{4(j+1)}$, and returns the median masked residual. It uses only acquired slices, no additional trained model, and zero neural calls. We compare it with adjacent-anchor MAE/RMSE, output prediction consistency (PC), and other frozen scores. The endpoint is examination-level $\times 4$ error or retrospective $\times 4 \rightarrow \times 2$ error reduction. Budgets are fixed at 10/20/30/50% and report oracle-normalized gain.

Independent units are 24 fastMRI examinations, 115 IXI examinations, and eight NingXia patients. Plane counts are descriptive. The mechanism chain is evaluated arrow by arrow: influence-to-alignment and influence-to-observable-fraction partial ranks control examination and geometry; alignment-to-normalized-disagreement additionally controls influence. Context-level ranking has only eight configurations and is reported separately. We use association language and do not claim causal mediation.

5. Results

5.1. Dependence exposes consensus blindness

Figure 2 shows the central intervention. Measurement-disjoint contexts rank error better than clean shared controls in every cohort. Disjoint-minus-shared $\Delta\rho$ is .092 [.043,.176] on fastMRI, .095 [.078,.126] on IXI, and .193 [.111,.371] on NingXia. Clean-sharing error cosines are .949, .968, and .995, leaving only 2.61%, 1.58%, and .25% of paired error energy observable. Under thresholds fixed from the disjoint condition, sharing raises high-error/low-disagreement prevalence by 27.1, 20.9, and 28.2 percentage points, respectively; every clustered interval excludes zero.

The effect is not tied to one learned estimator. Disjoint

case-level error-ranking ρ is .961/.955 for HAT/SwinIR on fastMRI and .910/.903 on NingXia. Analytical linear/cubic interpolation gives .977/.974 and .958/.967 on the same cohorts. IXI supplies the largest clean-control replication rather than a post-hoc favorable subset.

Table 1 separates effect size from sample size. The smallest cohort produces the largest point contrast but also the widest patient-clustered interval; it is therefore a boundary replication rather than the primary source of precision. The two brain cohorts independently support nearly identical $\Delta\rho$ despite different sample size and provenance.

5.2. Influence refines and survives falsification of count

Raw count has the expected overall relationship with error geometry, but the exact compact ladder is not monotone. Across all cohorts, overlap-count trends are $+.9$ for error cosine/shared fraction and $-.9$ to -1.0 for observable fraction/disagreement; nevertheless, the zero-to-one transition reverses. The early claim that every added common anchor must reduce observability is therefore rejected.

Analytical influence resolves this exception. Common coefficient mass correlates with error alignment at .815, .789, and .855 on fastMRI, IXI, and NingXia, versus .503, .466, and .630 for count. Across alignment, shared fraction, observable fraction, and disagreement, every clustered interval for the absolute-correlation advantage of influence over count is above zero (.189–.283).

The nonlinear result is consistent across two architectures and three broad context grids (Fig. 3). For error alignment, neural influence/count correlations are .838/.799 and .851/.796 on fastMRI, .797/.714 and .787/.712 on IXI, and .786/.743 and .772/.735 on NingXia for HAT/SwinIR. Across the 18 primary alignment/shared/observable endpoints, advantages range from .036 to .089 and every clustered interval excludes zero. Geometry- and examination-adjusted residual associations remain $|\rho| = .185$–$.395$.

The exact-one-overlap subset is the decisive beyond-count boundary. After controlling span, asymmetry, and examination, neural influence retains the expected association for every endpoint/model on fastMRI and IXI ($|\rho| = .135$–$.587$; intervals exclude zero). NingXia does not replicate it: error-alignment partial ρ is .014 for HAT and $-.039$ for SwinIR, all endpoint intervals cross zero, and some directions reverse. Thus the larger fastMRI/IXI cohorts support continuous influence beyond count; the eight-patient cohort supports only the broad descriptor and limits universal wording.

5.3. The local mechanism is stronger than a global ranking law

The arrow-wise analysis supports the local chain in all six strata. Influence-to-error-alignment partial ρ is .219–.410; alignment-to-normalized-disagreement is $-.933$ to $-.889$; influence-to-observable-fraction is $-.412$ to $-.211$. All 18 intervals exclude zero. The context-level consequence is weaker: greater alignment predicts poorer disagreement–error ranking in all strata, but intervals exclude zero in only three of six. A local pair can have aligned errors because common anchors influence both predictions, while the global case-ranking effect also depends on anatomy, error magnitude, and the small set of contexts. This is evidence for a local mechanism, not causal mediation or a universal ranking theorem.

A prespecified stability audit strengthens this interpretation. All 42 amplitude, sign/field, and direction-count strata pass the rank-stability gate (minimum $\rho = .942$). Physical-plane rank stability is .779–.954 and only one of three plane-pair gates passes. The response statistic is numerically stable at a declared target/plane but not a plane-invariant global property.

5.4. Architecture-family falsification

Figure 4 separates the robust local endpoint from the heterogeneous ranking consequence. Under the same fastMRI target, mask, model, and $\times 4$ physical grids, disjoint-minus-shared observable-error gain is 17.8 points for HAT, 17.6 for SwinIR, 8.0 for I3Net, 13.7 for SAINT, and 22.6 for CycleINR; all clustered intervals exclude zero. The equal-weight four-family estimate is 15.5 points [14.8,16.2]. Leave-one-family-out estimates remain 13.1–18.0 points and every case-level difference is positive. Measurement dependence therefore changes local error geometry across every evaluated paradigm.

The ranking arrow does not generalize uniformly. Disjoint-minus-shared error-ranking $\Delta\rho$ is .342 [.113,.711] for CycleINR, .018 [$-.061$,.107] for I3Net, $-.010$ [$-.143$,.114] for SAINT, and .057/.062 with intervals crossing zero for HAT/SwinIR. This directly falsifies a universal "disjoint improves ranking" statement while preserving the local alignment/observability mechanism.

The influence probe also identifies a boundary rather than a clean five-model replication. SAINT passes all amplitude/sign gates (minimum $\rho = .940$); its broad influence association is .706 [.680,.734] with error cosine and $-.698$ [$-.730$,$-.666$] with observable fraction. Yet influence exceeds count for error cosine by only .014 [$-.002$,.029], and its exact-one-overlap, geometry/examination-adjusted estimate is .063 [$-.021$,.149]. I3Net and CycleINR fail the preregistered stability gate and are classified as phenomenon-only. Hence stable non-transformer influence is directionally consistent, but "influence beats count across all architectures" is falsified.

Architecture diversity is not synonymous with independent failure. At fastMRI $\times 4$, the median examination error-rank correlation among the five models is .981, and six of

Table 1. Controlled measurement dependence. Stable-wrong Δ is the increase in high-error/low-disagreement prevalence under thresholds frozen from the disjoint condition.

Cohort	Independent units	ρ_{disjoint}	ρ_{shared}	$\Delta\rho$ [95% CI]	Observable error / stable-wrong Δ
fastMRI	24 examinations	.959	.867	.092 [.043,.176]	2.61% / +27.1 pp
IXI	115 examinations	.981	.886	.095 [.078,.126]	1.58% / +20.9 pp
NingXia	8 patients (22 series)	.892	.698	.193 [.111,.371]	.25% / +28.2 pp

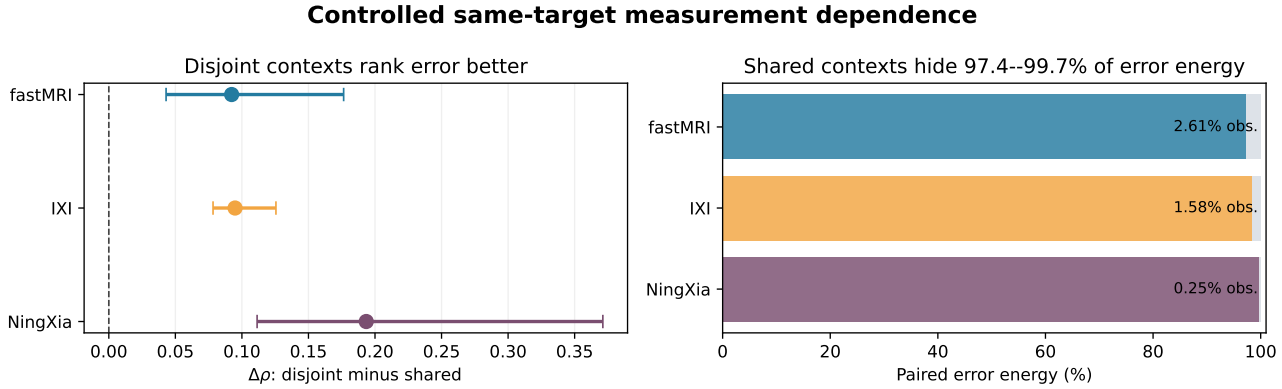


Figure 2. Same-target controlled intervention. Left: error-ranking improvement of measurement-disjoint over clean dependence-preserving contexts, with clustered 95% intervals. Right: shared (colored) and observable (gray) paired-error energy under clean sharing. NingXia uncertainty is patient-clustered over eight patients.

24 cases are in the top error quartile for all five. Pixelwise error cosine is lower for CycleINR–I3Net (.564 median) than HAT–SwinIR (.918), showing that local error diversity can increase even while case difficulty remains shared.

5.5. Practical consequence without superiority

Figure 5 evaluates acquired-anchor cross-validation as a consequence, not the central novelty. Anchor-CV ranks $\times 4$ examination error with $\rho = .943/.955/.958$ on fastMRI/IXI/NingXia. Adjacent MAE gives .934/.956/.973 and adjacent RMSE .942/.956/.959; paired differences are not consistently significant. Conditional association beyond adjacent MAE, scale, and spacing is $.333/-.059/-.100$. Anchor-CV is therefore an interpretable acquired-only difficulty signal, not unique information.

The retrospective densification endpoint remains practically meaningful. Anchor-CV correlates with $\times 4 \rightarrow \times 2$ benefit at .871 on fastMRI and .948 on IXI. On IXI its oracle-normalized gains are .920/.936/.972/.973 at the four frozen budgets. Adjacent MAE and output PC are comparable, and output PC reaches .970 benefit correlation and .996 gain at the 50% IXI budget. The practical claim is availability and low cost: Anchor-CV requires already acquired anchors, zero dense targets at scoring, zero additional neural calls, and 0.42–3.97 ms CPU in the measured cases. It does not dominate every score.

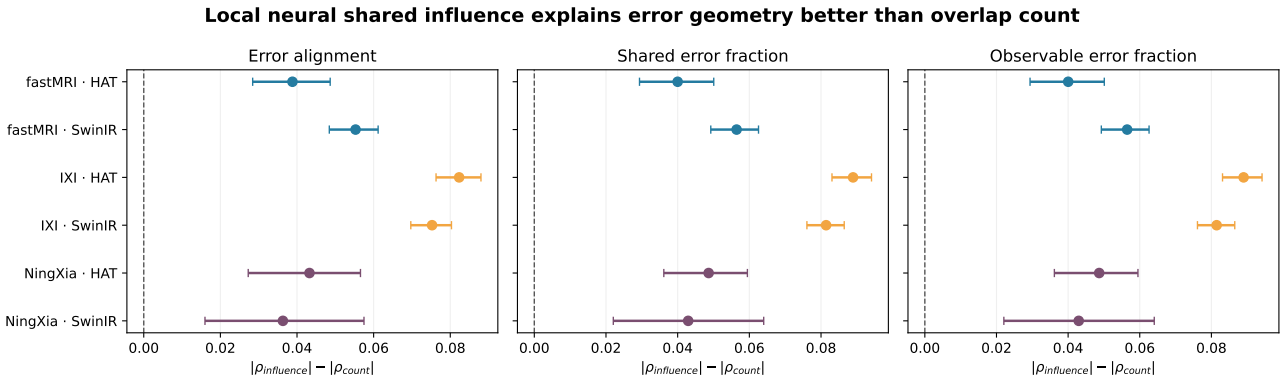


Figure 3. Neural shared influence improves absolute association over raw anchor-overlap count for error alignment, shared error fraction, and observable error fraction in all six cohort–architecture strata. Points show $|\rho_{\text{influence}}| - |\rho_{\text{count}}|$ with examination/patient-clustered 95% intervals.

Table 2. Broad-grid and fixed-one-overlap neural evidence. Broad columns compare error-alignment correlation for count and neural influence; fixed columns give geometry/examination-adjusted influence correlation and clustered intervals.

Cohort	Model	Broad ρ_{count}	Broad $\rho_{\text{influence}}$	Advantage [95% CI]	Fixed-one-overlap partial ρ [95% CI]
fastMRI	HAT	.799	.838	.039 [.028,.049]	.169 [.053,.283]
	SwinIR	.796	.851	.055 [.048,.061]	.350 [.212,.464]
IXI	HAT	.714	.797	.082 [.076,.088]	.346 [.259,.424]
	SwinIR	.712	.787	.075 [.070,.080]	.258 [.181,.335]
NingXia	HAT	.743	.786	.043 [.027,.057]	.014 [−.232,.223]
	SwinIR	.735	.772	.036 [.016,.057]	−.039 [−.175,.144]

Table 3. Mechanism-chain partial Spearman ranges across HAT/SwinIR. Every local-arrow interval excludes zero. The final column counts architecture strata with a context-ranking interval excluding zero.

Cohort	Influence→alignment	Alignment→norm. disagreement	Influence→observable	Alignment→ranking
fastMRI	.372–.410	−.933–−.926	−.412–−.370	−.333–−.143 (0/2)
IXI	.321–.347	−.893–−.890	−.342–−.298	−.643–−.500 (2/2)
NingXia	.219–.258	−.898–−.889	−.231–−.211	−.667–−.548 (1/2)

Table 4. Practical consequence and strong baselines. Benefit is the retrospective $\times 4 \rightarrow \times 2$ error reduction. Dashes indicate that NingXia is excluded from the frozen two-cohort consequence rule.

Cohort	Anchor-CV ρ vs. x4 error	Adjacent MAE ρ	Anchor-CV ρ vs. benefit	Best competing benefit ρ
fastMRI	.943	.934	.871	.846 adjacent MAE / .843 PC
IXI	.955	.956	.948	.970 PC
NingXia	.958	.973	—	—

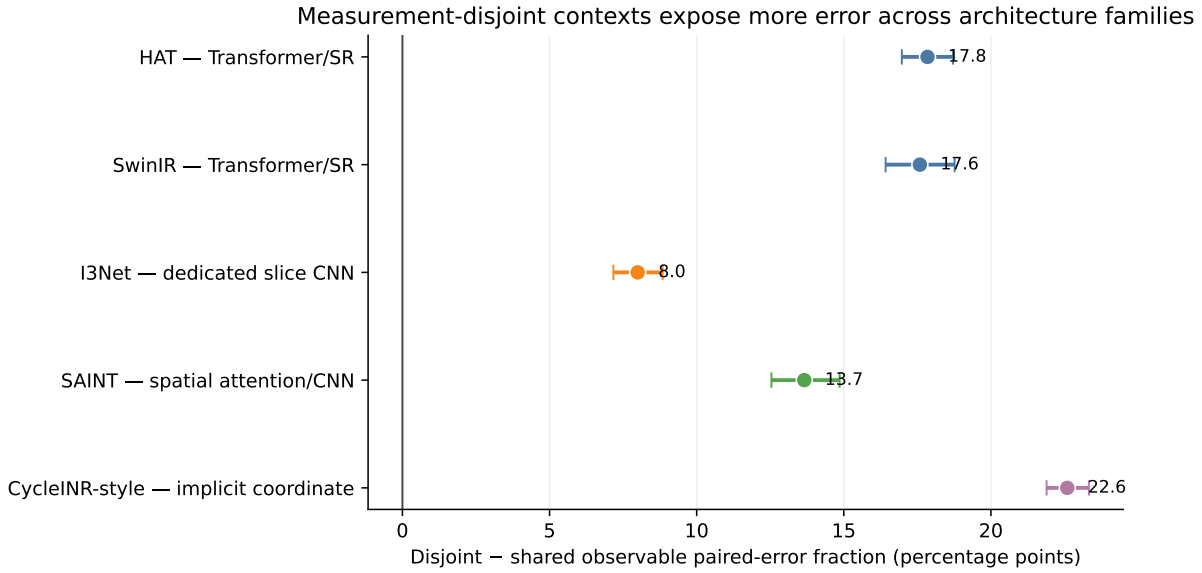


Figure 4. Preregistered architecture-family falsification. Disjoint-minus-shared observable paired-error fraction remains positive across transformer SR, inter/intra-slice CNN, spatial conditional CNN, and implicit-coordinate interpolation. The plot does not imply a universal examination-ranking gain.

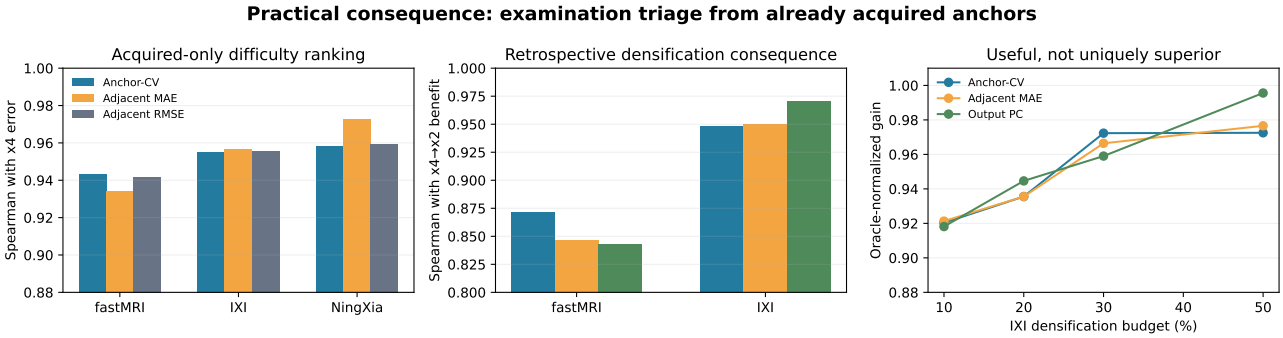


Figure 5. Practical consequence. Anchor-CV uses acquired anchors only and strongly ranks examination difficulty (left) and retrospective $\times 4 \rightarrow \times 2$ benefit (middle). The IXI allocation curves (right) show large gain above random but no universal winner. Output PC requires existing reconstruction outputs; Anchor-CV requires none.

6. Discussion

The reviewer-memory message is simple: *disagreement is only as informative as the independence of the information generating the compared predictions*. MRI interpolation makes this testable because target position and physical anchor identity are known. When contexts depend strongly on the same measurements, their local responses and target errors align, causing disagreement to understate error. Measurement-disjoint contexts expose a larger error component.

The progression from overlap count to influence is scientifically important. Count is attractive because it is exact and acquisition-only, but the zero-to-one reversal falsifies a strict dose law. Unequal coefficient mass explains analytical configurations, and paired neural finite responses transfer the idea to nonlinear interpolators. Plane dependence is not a failed invariance theorem because the final claim is local: influence belongs to a target–context–input configuration. It would contradict a claim of one intrinsic global model coefficient; we make no such claim.

The result also clarifies what is and is not actionable. Anchor-CV offers an inexpensive acquired-only examination signal and tracks retrospective densification value across fastMRI and IXI. Yet adjacent differences carry essentially the same rank information, output PC can be stronger when outputs already exist, and neither score localizes the worst physical plane. The mechanism therefore informs reliability-context design and examination triage without implying calibrated uncertainty, autonomous reacquisition, or a universally superior scalar.

Several limitations are material. NingXia contains 22 series but only eight independent patients; its fixed-count result is null/mixed. Architecture breadth improves, but all evaluated data remain MRI T2 and the P0 controlled experiment is fastMRI-only. The local SAINT and CycleINR-style implementations are adapted/internal reproductions, not exact official models. Only SAINT passes all new-architecture influence-stability gates; its beyond-count estimate is imprecise. I3Net/CycleINR support the physical-dependence phenomenon but not a finite-response influence claim. The grids have different feasible overlap support, the final ranking arrow remains heterogeneous, and finite differences cannot identify a global Jacobian. The $\times 2$ consequence is retrospective; scanner time, motion, and clinical cost are absent. No result establishes diagnostic safety.

Evidence hierarchy. We treat fastMRI and IXI as the primary mechanism evidence because they provide 24 and 115 independent examinations. NingXia changes anatomy and site but contains eight patient clusters; it exposes heterogeneity rather than contributing artificial pooled precision. The same hierarchy is applied to favorable and unfavorable results: its broad influence direction is reported, and so is its exact fixed-count failure. This prevents both an underpowered nonreplication from erasing the primary effect and selective omission from overstating universality.

7. Conclusion

Deterministic MRI interpolators can agree while jointly wrong because prediction differences cannot observe their shared error. Controlled same-target interventions show that dependence-preserving contexts hide more error than disjoint contexts across three cohorts and five evaluated neural architectures. Architecture falsification strengthens the local physical-dependence–error-geometry claim but rejects two stronger versions: examination ranking does not always improve, and finite-response influence is not stable or superior to count for every architecture. Acquired-anchor cross-validation remains a lightweight consequence without being universally best. Reliability from disagreement depends on the independence of influential information, but that dependence must be measured and validated for the architecture at hand.

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